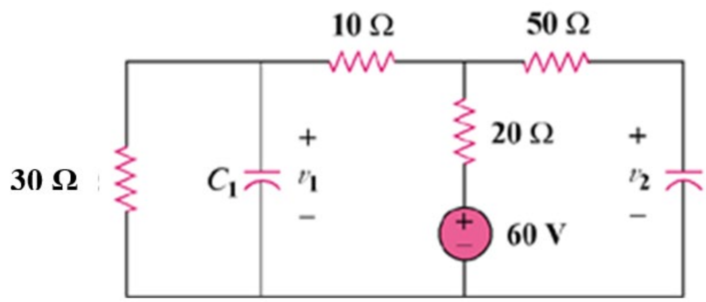
ENGR 2303: Electrical Science

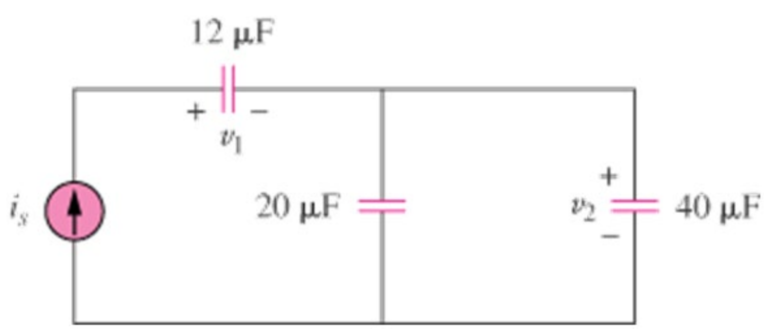
HW 6_Ch6

1) The current through a 0.5-F capacitor is $6(1-e^{-t})$ A. Determine the voltage and power at t=2 s. Assume v(0) = 0.

2) Find the voltage across the capacitors in the circuit below under dc conditions.

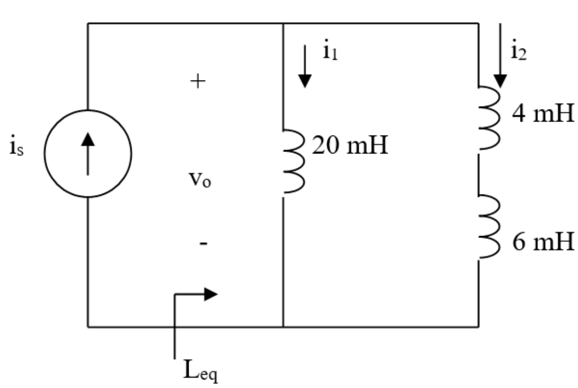


3) In the circuit below, let $i_s = 30e^{-2t}$ mA and $v_1(0) = 50$ V, $v_2(0) = 20$ V. Determine: (a) $v_1(t)$ and $v_2(t)$, (b) the energy in each capacitor at $t = 0.5$ s.



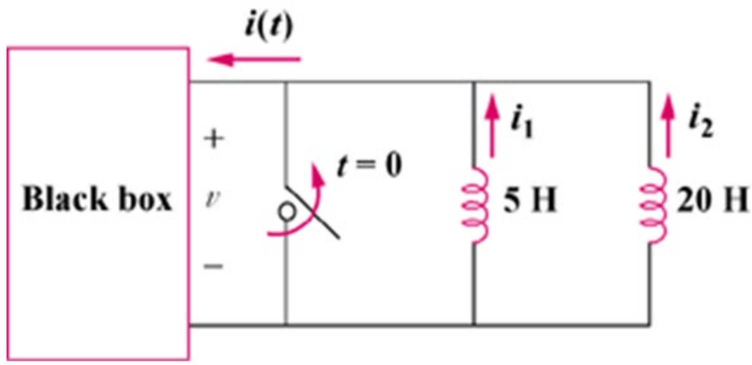
4) Consider the circuit below. Find: (a) L_{eq}, $i_1(t)$ and $i_2(t)$ if $i_s = 3e^{-t}$ mA, (b) $v_o(t)$,

(c) energy stored in the 20-mH inductor at t=1s.



5) The inductors in the circuit below are initially charged and are connected to the black box at $t = 0$. If $i_1(0) = 4$ A, $i_2(0) = -2$ A, and $v(t) = 50e^{-200t}$ mV, $t \geq 0$ find:

(a). the energy initially stored in each inductor,
(b). the total energy delivered to the black box from $t = 0$ to $t = \infty$,
(c). $i_1(t)$ and $i_2(t)$, $t \geq 0$,
(d). $i(t)$, $t \geq 0$.



6) An op amp differentiator has ***R***= 250 kΩ and ***C*** = 10 μF.

The input voltage is a ramp ***r(t)*** = 12 ***t*** mV. Find the output voltage.

7) Design an analog computer circuit to solve the following ordinary differential equation.

$$\frac{dy(t)}{dt} + 4y(t) = f(t) \qquad \text{where } y(0) = 1 \text{ V.}$$